

The Real Numbers

Classification

<i>Natural numbers (or positive integers)</i>	$1, 2, 3, \dots$
<i>Integers</i>	$0, 1, -1, 2, -2, 3, -3, \dots$
<i>Rational numbers</i>	$\{x : x = p/q, \text{ where } p, q \text{ are intergers and } q \neq 0\}. \dagger$
Examples	$\frac{2}{3}, \frac{-19}{7}, \frac{4}{1} (= 4).$
<i>Irrational numbers</i>	real numbers that are not rational numbers.
Examples	$\sqrt{2}, \sqrt[3]{7}, \pi$, the solutions of the equation $x^2 - 5 = 0$.

$\dagger$ Recall that division by 0 is undefined.

The Real Numbers

Decimal Representation

Each real number can be represented by a decimal. If $r = p/q$ is a rational number, then its decimal representation is found by dividing the denominator q into the numerator p . The resulting decimal expansion will either *terminate* or *repeat*. For example,

$$\frac{3}{5} = 0.6, \quad \frac{27}{20} = 1.35, \quad \text{and} \quad \frac{43}{8} = 5.375$$

are terminating decimals and

$$\frac{2}{3} = 0.6666 \dots = 0.\overline{6}, \quad \frac{15}{11} = 1.363636 \dots = 1.\overline{36}, \quad \text{and} \quad \frac{116}{37} = 3.135135 \dots = 3.\overline{135}$$

are repeating decimals (the bar over the sequence of digits indicates that the sequence repeats indefinitely). The converse is also true; every terminating or repeating decimal represents a rational number.

The decimal expansion of an irrational number neither terminates nor repeats. For example, the numbers

$$\sqrt{2} = 1.414213562 \dots, \quad \pi = 3.141592653 \dots \quad \text{and} \\ r = 0.10110111011110 \dots$$

are irrational numbers.

If we stop the decimal expansion of a given number at a certain decimal place, then the result is a rational number which approximates the given number. For instance, $1.414 = 1414/1000$ and $3.14 = 314/100$ are common rational number approximations for $\sqrt{2}$ and π , respectively. More accurate approximations can be obtained by taking more decimal places in the expansions.

The Real Numbers

Geometric Representation

A fundamental concept in mathematics which connects the abstract notion of *real number* with the geometric notion of *point* is the representation of the real numbers as points on a straight line. This is done by choosing an arbitrary point O on a horizontal line to represent the number 0, and another point U (usually taken to the right of O) to represent the number 1. See Figure 1.2.1. The point O is called the *origin*, and the distance between O and U determines a scale (a unit length). With O and U specified, each real number can be represented as a point on the line and, conversely, each point on the line represents a real number. A line representing the real numbers is called a *number line* or *real line*; the number associated with a point P on the line is called the *coordinate* of P . The *positive* numbers are identified with the points on the right side of O and the *negative* numbers with the points to the left of O . The point representing the number a ($a \neq 0$) is a units from O if a is positive and $-a$ units from O if a is negative. In this context, we will frequently refer to real numbers as “points,” and by this we will mean “points on a number line.” Figure 1.2.2 shows some numbers plotted as points on a number line.

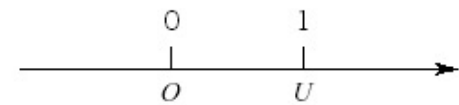


Figure 1.2.1

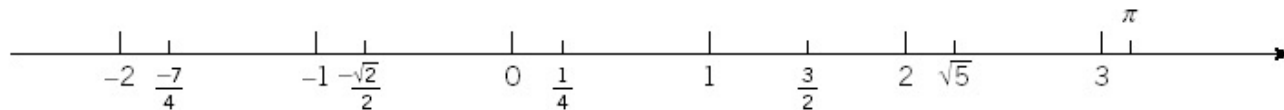


Figure 1.2.2

The Real Numbers

Order Properties

If a and b are real numbers, then a is less than b , denoted $a < b$, if $b - a$ is a positive number. This is equivalent to saying that b is greater than a , which is denoted $b > a$. Geometrically, $a < b$ if the point a lies to the left of the point b on a number line. The notation $a \leq b$ means that either $a < b$ or $a = b$ (equivalently, $b \geq a$).

The real numbers are *ordered* in the sense that if a and b are real numbers, then exactly one of the following holds:

$$a < b, \quad a = b, \quad a > b. \quad (\text{trichotomy})$$

The symbols $<$, $>$, $\leq$, $\geq$ are called *inequalities*. Inequalities satisfy the following properties:

- (i) If $a < b$ and $b < c$, then $a < c$. (transitive property)
- (ii) If $a < b$, then $a + c < b + c$ for all real numbers c .
- (iii) If $a < b$ and $c < d$, then $a + c < b + d$.
- (iv) If $a < b$ and $c > 0$, then $ac < bc$.
- (v) If $a < b$ and $c < 0$, then $ac > bc$.

The corresponding properties hold for $>$, $\leq$, $\geq$. A crucial point is Property (v): if an inequality is multiplied by a negative quantity, then the “direction” of the inequality is reversed. Techniques for solving inequalities use these properties and are reviewed in Section 1.3.

The Real Numbers

Absolute Value

Two important properties of a real number a are its *sign* and its *size*, or *magnitude*. Geometrically, the sign of a tells us whether the point a is on the right or left of 0 on a number line. The magnitude of a is the distance between the point a and 0; 0 itself does not have a sign and its magnitude is 0. The magnitude of a is more commonly called *absolute value of a* , denoted $|a|$. The absolute value of a is also given by

$$|a| = \begin{cases} a, & \text{if } a \geq 0 \\ -a, & \text{if } a < 0. \end{cases}$$

Other characterizations $|a| = \max\{a, -a\}; |a| = \sqrt{a^2}.$

Geometric interpretations $|a| = \text{distance between } a \text{ and } 0;$

$$|a - c| = \text{distance between } a \text{ and } c.$$

Other properties (i) $|a| = 0$ iff $a = 0$.[†]

(ii) $|-a| = |a|.$

(iii) $|ab| = |a||b|.$

(iv) $|a + b| \leq |a| + |b|$ (the triangle inequality).^{††}

(v) $||a| - |b|| \leq |a - b|$ (a variant of the triangle inequality).

(vi) $|a|^2 = |a^2| = a^2$

Techniques for solving inequalities involving absolute values are also reviewed in Section 1.3.

[†] By “iff” we mean “if and only if.” This expression is used so often in mathematics that it’s convenient to have an abbreviation for it.

^{††} The absolute value of the sum of two numbers cannot exceed the sum of their absolute values. This is analogous to the fact that in a triangle the length of one side cannot exceed the sum of the lengths of the other two sides.

The Real Numbers

Intervals

Suppose that $a < b$. The *open interval* (a, b) is the set of all numbers between a and b :

$$(a, b) = \{x : a < x < b\}.$$



The *closed interval* $[a, b]$ is the open interval (a, b) together with the endpoints a and b :

$$[a, b] = \{x : a \leq x \leq b\}.$$



There are seven other types of intervals:

$$(a, b] = \{x : a < x \leq b\},$$

$$[a, b) = \{x : a \leq x < b\},$$

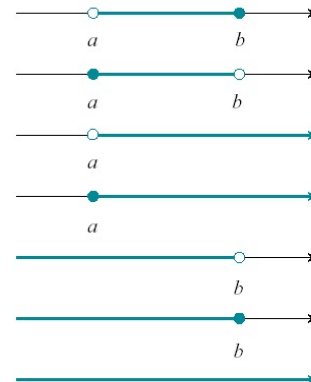
$$(a, \infty) = \{x : a < x\},$$

$$[a, \infty) = \{x : a \leq x\},$$

$$(-\infty, b) = \{x : x < b\},$$

$$(-\infty, b] = \{x : x \leq b\},$$

$$(-\infty, \infty) = \text{the set of real numbers}.$$



Interval notation is easy to remember: we use a square bracket to include an endpoint and a parenthesis when the endpoint is not included. On a number line, the inclusion or exclusion of an endpoint is signified with a solid “dot” or an open “dot,” respectively. The symbols ∞ and $-\infty$, read “infinity” and “negative infinity” (or “minus infinity”), are not real numbers. The ∞ symbol in the intervals given above indicates that the interval extends indefinitely in the positive direction; the $-\infty$ indicates that the interval extends indefinitely in the negative direction. Since ∞ and $-\infty$ are not real numbers, we do not have intervals of the form $[a, \infty]$ or $[-\infty, b)$, and so forth.

Algebra

Factorials

Let n be a positive integer. Then n *factorial*, denoted $n!$, is the product of the integers from 1 to n . Thus,

$$1! = 1, \quad 2! = 1 \cdot 2, \quad 3! = 1 \cdot 2 \cdot 3, \quad \dots, \quad n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1) \cdot n.$$

Note that

$$1! = 1, \quad 2! = 2, \quad 3! = 6.$$

You can verify that $4! = 24$, $5! = 120$, and so on. Finally, to ensure that certain formulas will hold for all *nonnegative* integers, we define $0! = 1$

Algebra

Quadratic Equations

The roots of a quadratic equation

$$ax^2 + bx + c = 0 \quad \text{with } a \neq 0$$

are given by the quadratic formula

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

If $b^2 - 4ac > 0$, the equation has two real roots:

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

In this case, the quadratic expression $ax^2 + bx + c$ can be written in the factored form:

$$ax^2 + bx + c = a(x - r_1)(x - r_2).$$

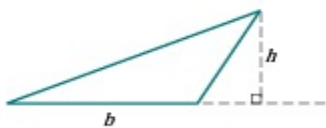
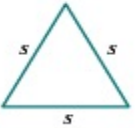
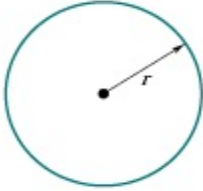
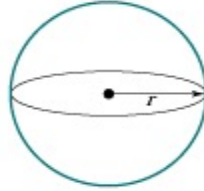
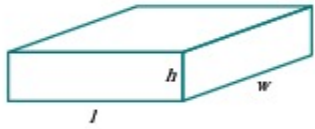
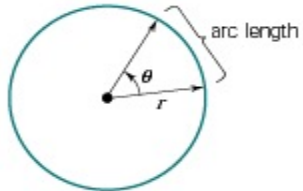
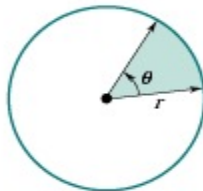
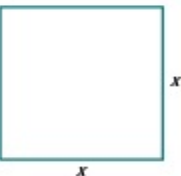
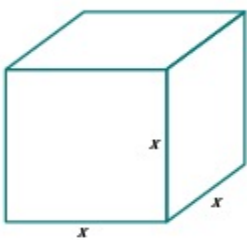
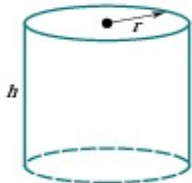
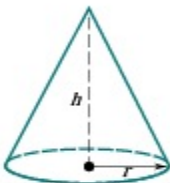
If $b^2 - 4ac = 0$, then $r_1 = r_2$, and the equation has only one real root $r = -b/2a$. Here, the quadratic can be written in the factored form:

$$ax^2 + bx + c = a(x - r)^2.$$

If $b^2 - 4ac < 0$, the equation has no real roots and the quadratic cannot be factored.

To find the roots of a quadratic equation one typically tries to factor the quadratic expression $ax^2 + bx + c$ into one of the forms $a(x - r_1)(x - r_2)$ or $a(x - r)^2$. If this turns out to be difficult, or impossible, then the quadratic formula is used.

Geometry

Triangle  area = $\frac{1}{2}bh$	Equilateral Triangle  area = $\frac{1}{4}\sqrt{3}s^2$	Circle  area = πr^2 circumference = $2\pi r$	Sphere  volume = $\frac{4}{3}\pi r^3$ surface area = $4\pi r^2$
Rectangle  area = lw perimeter = $2l + 2w$ diagonal = $\sqrt{l^2 + w^2}$	Rectangular Solid  volume = lwh surface area = $2lw + 2lh + 2wh$	Sector of a Circle: radius r, central angle θ measured in radians (see Section 1.6).  arc length = $r\theta$  area = $\frac{1}{2}r^2\theta$	
Square  area = x^2 perimeter = $4x$ diagonal = $x\sqrt{2}$	Cube  volume = x^3 surface area = $6x^2$	Right Circular Cylinder  volume = $\pi r^2 h$ lateral area = $2\pi rh$ total surface area = $2\pi r^2 + 2\pi rh$	Right Circular Cone  volume = $\frac{1}{3}\pi r^2 h$ slant height = $\sqrt{r^2 + h^2}$ lateral area = $\pi r\sqrt{r^2 + h^2}$ total surface area = $\pi r^2 + \pi r\sqrt{r^2 + h^2}$

Figure, pp.9, 10

Inequalities & Absolute value

Recall that for each real number a

$$(1.3.1) \quad |a| = \begin{cases} a & \text{if } a \geq 0, \\ -a, & \text{if } a < 0, \end{cases} \quad |a| = \max\{a, -a\}, \quad |a| = \sqrt{a^2}$$

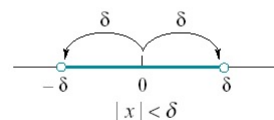
We begin with the inequality

$$|x| < \delta,$$

where δ is some positive number. To say that $|x| < \delta$ is to say that x lies within δ units of 0 or, equivalently, that x lies between $-\delta$ and δ . Thus

(1.3.2)

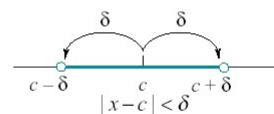
$$|x| < \delta \quad \text{iff} \quad -\delta < x < \delta.$$



To say that $|x - c| < \delta$ is to say that x lies within δ units of c or, equivalently, that x lies between $c - \delta$ and $c + \delta$. Thus

(1.3.3)

$$|x - c| < \delta \quad \text{iff} \quad c - \delta < x < c + \delta.$$



Somewhat more delicate is the inequality

$$0 < |x - c| < \delta.$$

Here we have $|x - c| < \delta$ with the additional requirement that $x \neq c$. Consequently

(1.3.4)

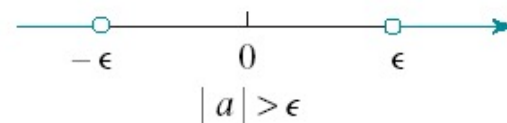
$$0 < |x - c| < \delta \quad \text{iff} \quad c - \delta < x < c \quad \text{or} \quad c < x < c + \delta.$$

Inequalities & Absolute value

Let $\epsilon > 0$. If you think of $|a|$ as the distance between a and 0, then

(1.3.5)

$$|a| > \epsilon \quad \text{iff} \quad a > \epsilon \quad \text{or} \quad a < -\epsilon.$$



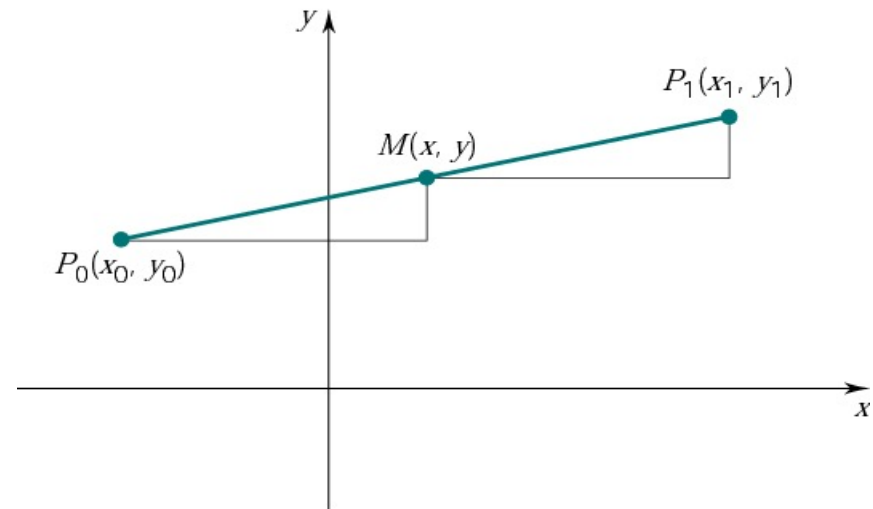
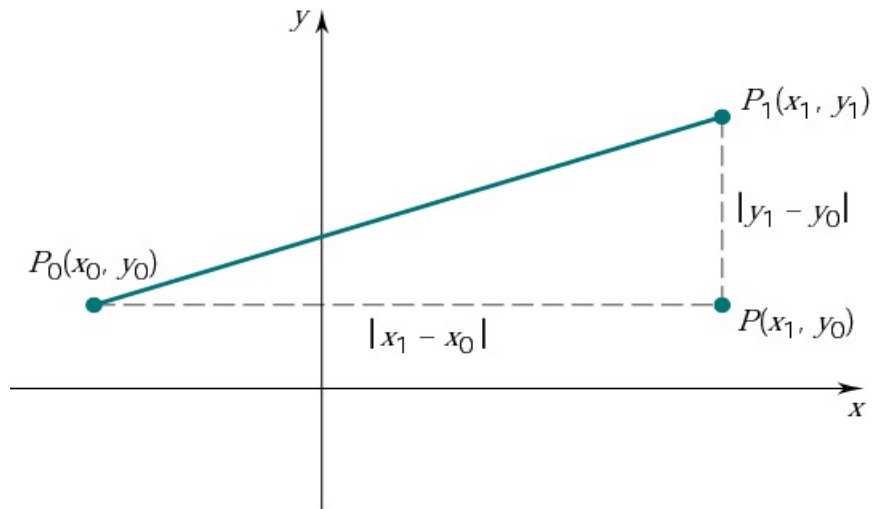
We come now to one of the fundamental inequalities of calculus: for all real numbers a and b ,

(1.3.6)

$$|a + b| \leq |a| + |b|.$$

This is called the *triangle inequality* in analogy with the geometric maxim “in a triangle the length of each side is less than or equal to the sum of the lengths of the other two sides.”

Distance and Midpoint Formulas



Lines

Lines

Let l be a straight line in the plane, and let $P_0(x_0, y_0)$ and $P_1(x_1, y_1)$ be any two distinct points on l .

(i) Slope The *slope* of l is given by

$$m = \frac{y_1 - y_0}{x_1 - x_0} \quad \text{provided } x_1 \neq x_0.$$

Also, $m = \tan \theta$, †

where θ , called the *inclination* of l , is the angle between the positively directed x -axis and l measured in the counterclockwise direction (see Figure 1.4.5). If $x_1 = x_0$, then l is a vertical line; its slope is undefined and $\theta = \pi/2 = 90^\circ$.

Lines

(iii) Equations An equation for l is an equation which involves two variables, say x and y , and has the property that a point $P(x_0, y_0)$ lies on l iff its coordinates, x_0 and y_0 , satisfy the equation. The standard forms are as follows:

vertical line $x = a.$

horizontal line $y = b.$

point-slope form $y - y_0 = m(x - x_0).$

two-point form $y - y_0 = \frac{y_1 - y_0}{x_1 - x_0}(x - x_0).$

slope-intercept form $y = mx + b.$ $(y = b \text{ when } m = 0)$

two-intercept form $\frac{x}{a} + \frac{y}{b} = 1.$

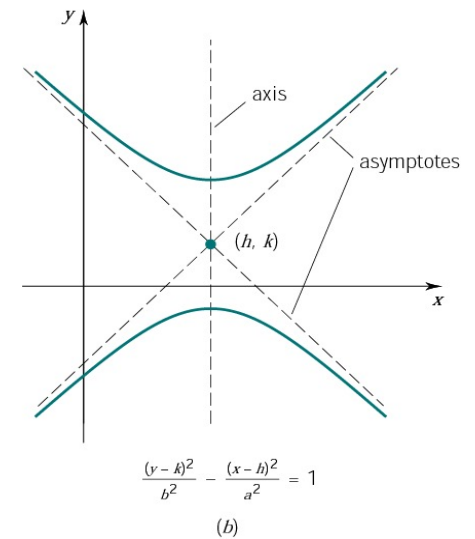
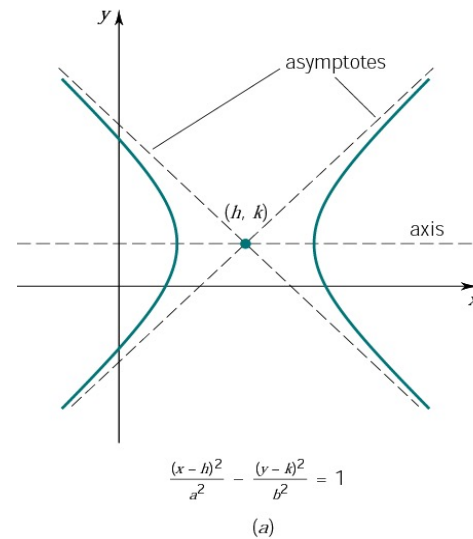
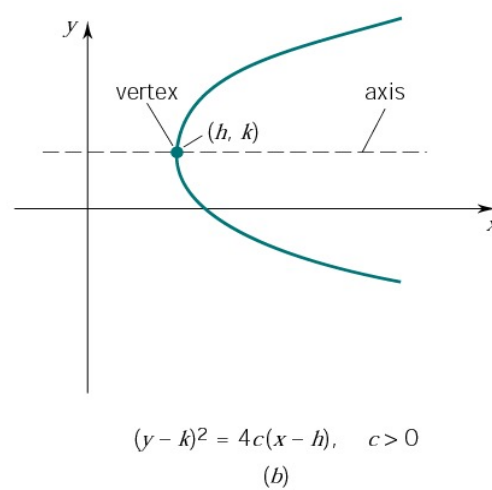
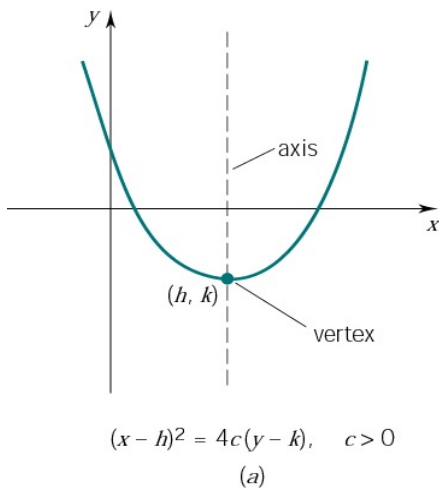
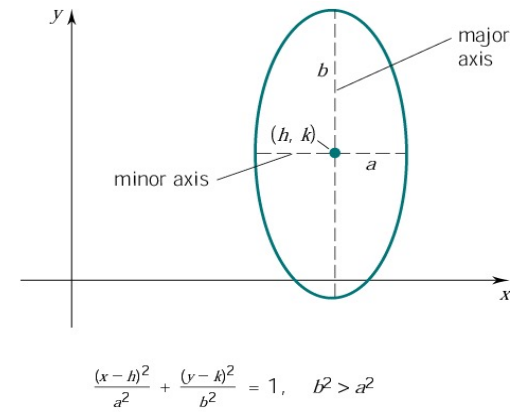
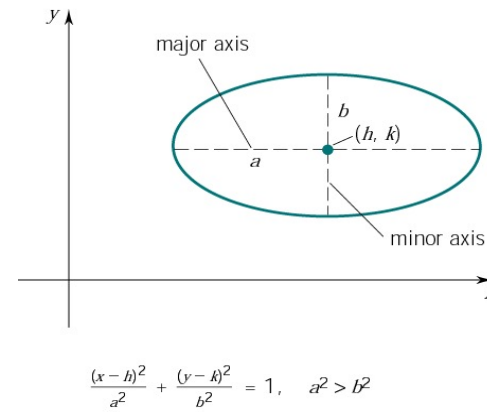
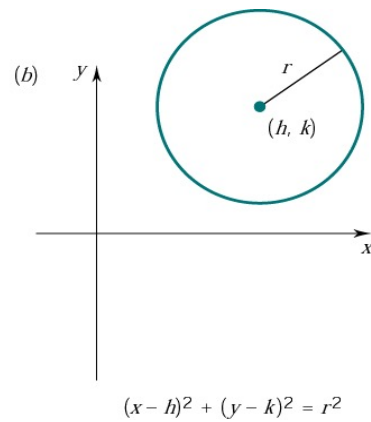
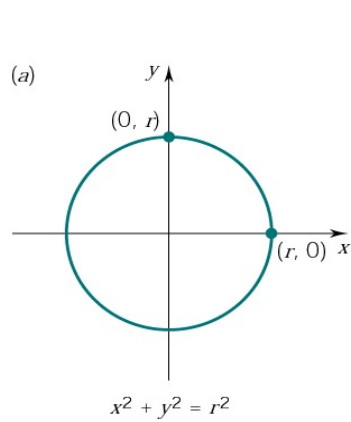
general form $Ax + By + C = 0.$ $(A \text{ and } B \text{ not both } 0)$

Lines

(iv) Parallel and Perpendicular Lines Given two nonvertical lines l_1 and l_2 with slopes m_1 and m_2 , respectively, then l_1 and l_2 are

$$\begin{array}{ll} \text{parallel} & \text{iff } m_1 = m_2, \\ \text{perpendicular} & \text{iff } m_1 m_2 = -1. \end{array}$$

Conic Sections



Circle, ellipse, parabola, hyperbola, pp. 23, 24, 25

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Conic Sections

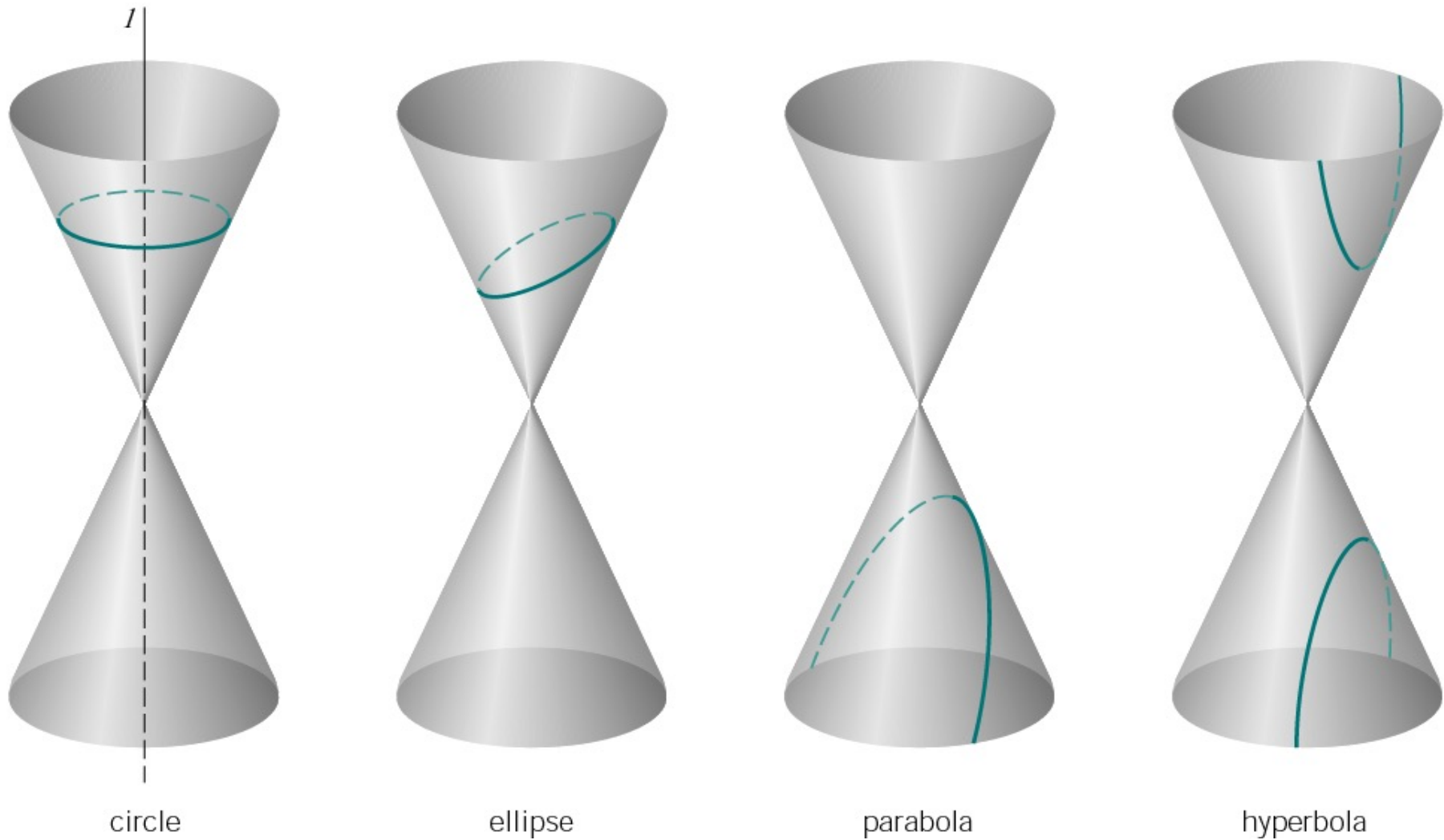
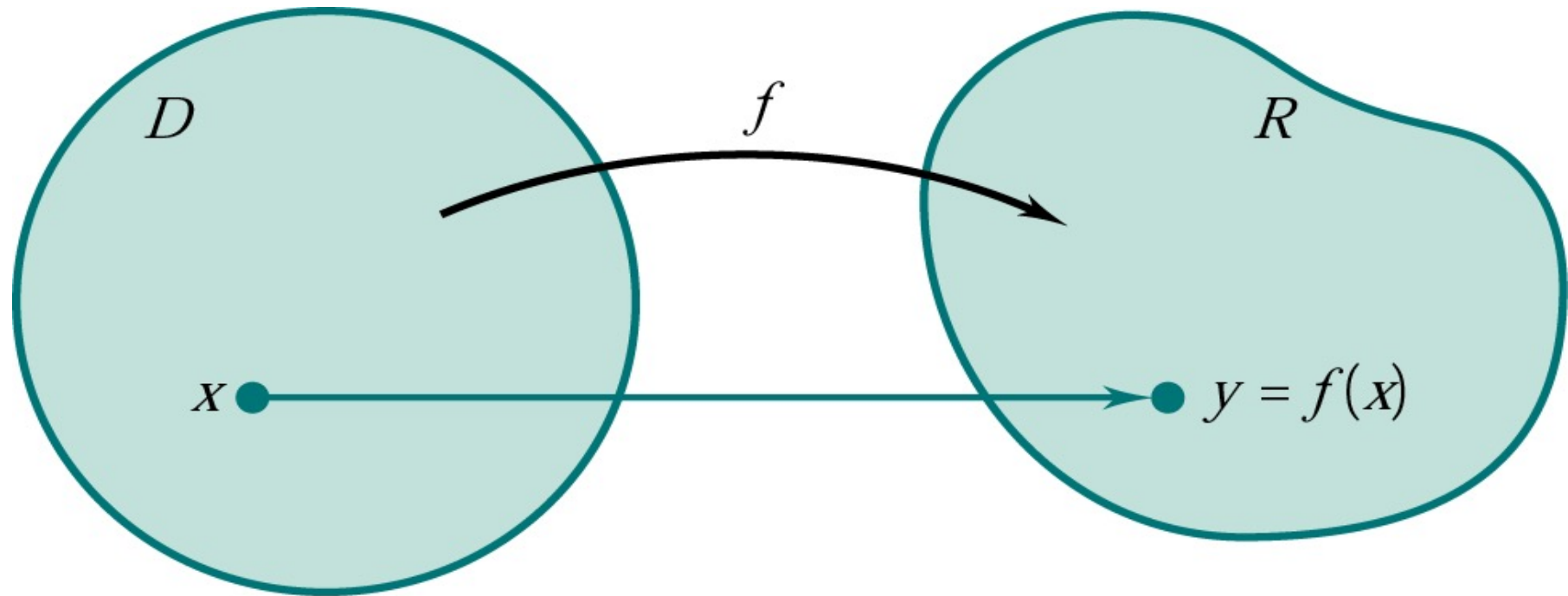


Figure 1.4.13

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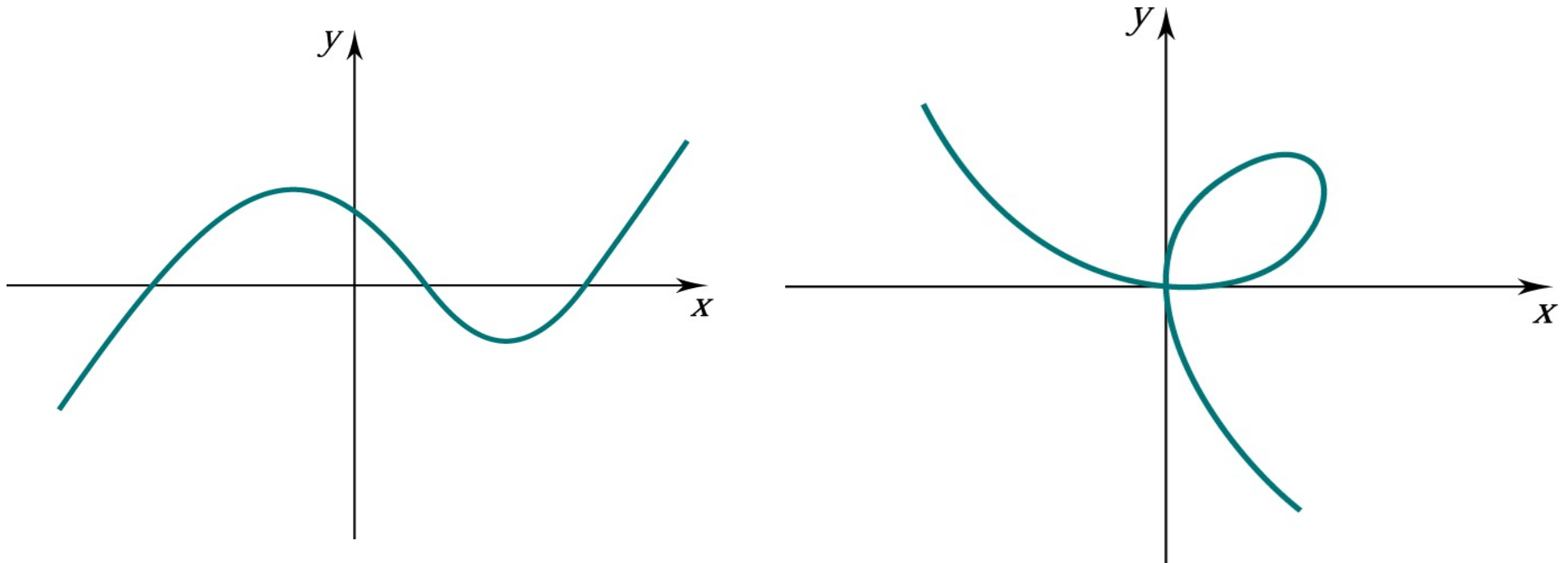
Functions



Domain and range, p.28, figure 1.5.1

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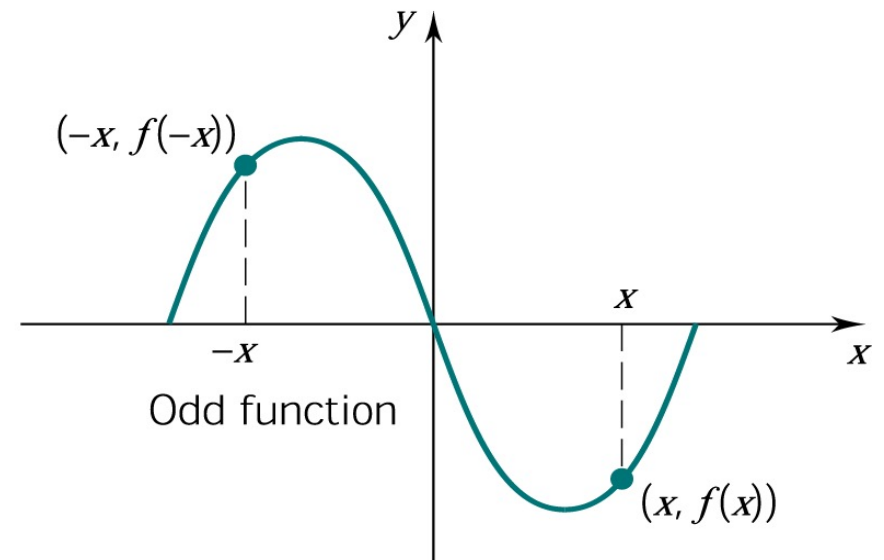
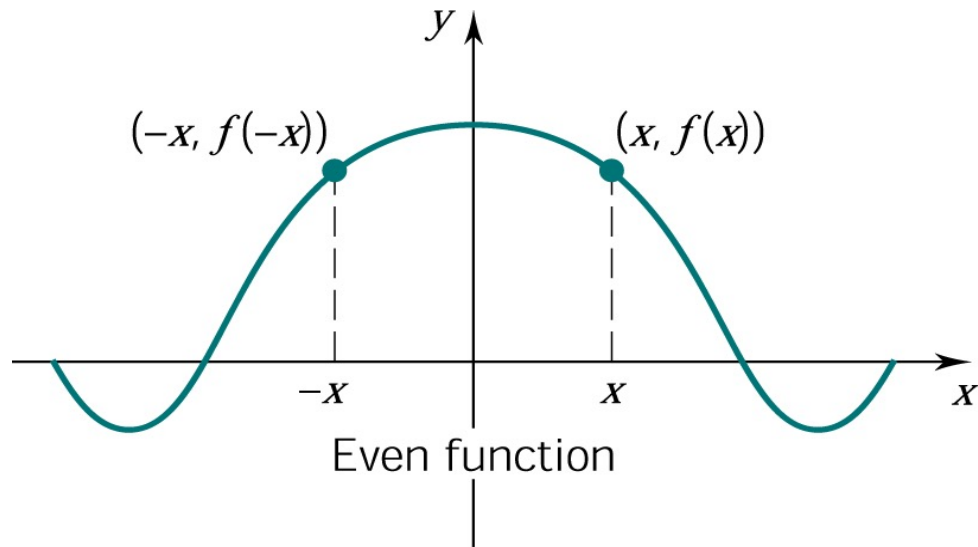
Functions



Vertical line test, p.31, figures 1.5.6-7

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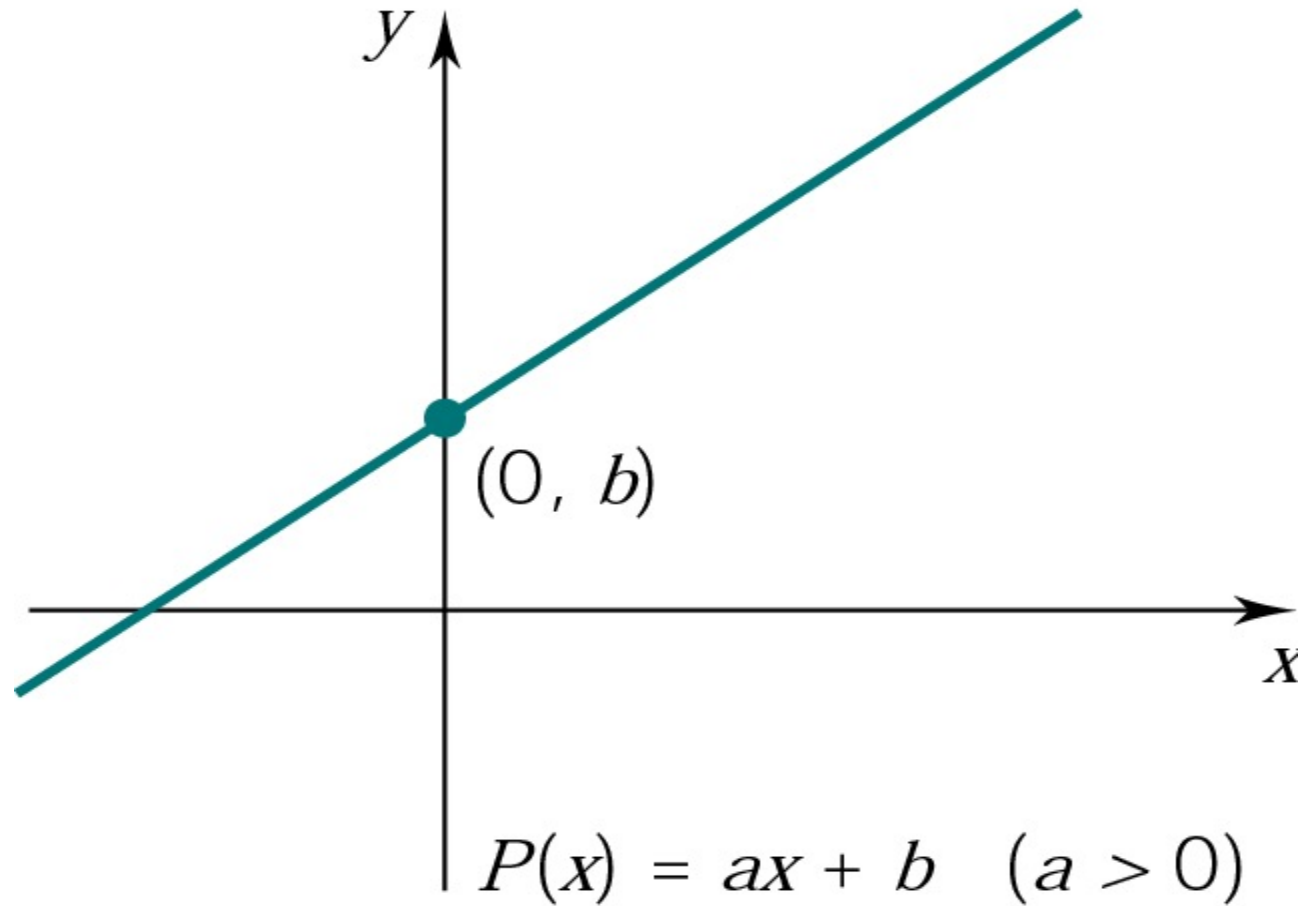
Functions



Even & odd functions, symmetry, p.31, figures 1.5.8-9

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The Elementary Functions



The Elementary Functions

The quadratic functions

$$P(x) = ax^2 + bx + c, \quad a \neq 0$$



$a > 0$

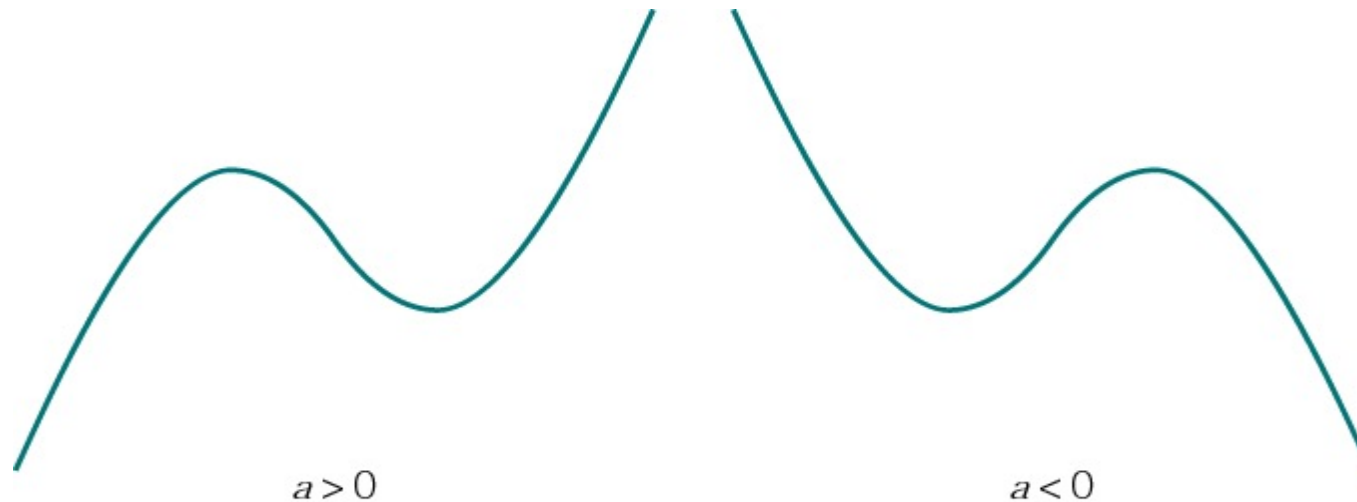


$a < 0$

The Elementary Functions

Polynomials of degree 3 have the form $P(x) = ax^3 + bx^2 + cx + d, a \neq 0$. These functions are called *cubics*. In general, the graph of a cubic has one of the two following shapes, again determined by the sign of a (Figure 1.6.4). Note that we have not tried to locate these graphs with respect to the coordinate axes. Our purpose here is simply to indicate the two typical shapes. You can see, however, that for a cubic there are three possibilities: 3 real roots, 2 real roots, or 1 real root.

Polynomials become more complicated as the degree increases. In Chapter 4 we show how to use calculus to analyze polynomials of higher degree and to sketch their graphs.



The Elementary Functions

Rational Functions

A *rational function* is a function of the form

$$R(x) = \frac{P(x)}{Q(x)},$$

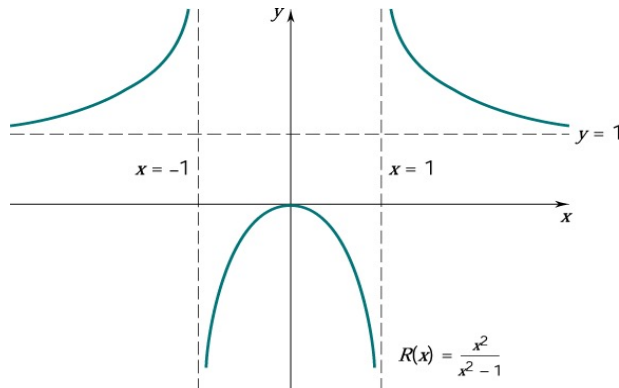
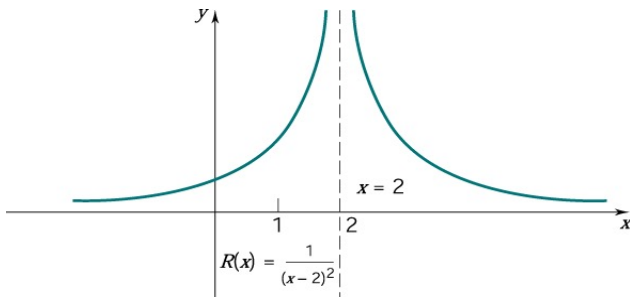
where P and Q are polynomials. Note that every polynomial P is a rational function: $P(x) = P(x)/1$ is the quotient of two polynomials. Since division by 0 is not allowed, a rational function $R = P/Q$ is not defined at those points x (if any) where $Q(x) = 0$; R is defined at all other points. Thus, $\text{dom}(R) = \{x : Q(x) \neq 0\}$.

Rational functions are more difficult to analyze and more difficult to graph than polynomials. For example, it is necessary to analyze the behavior of a rational function $R = P/Q$ in the vicinity of a zero of Q , as well as the behavior of R for large values of x , both positive and negative. If P and Q have no common factors (P/Q is in “lowest terms”), then the zeros of Q correspond to *vertical asymptotes* of the graph of R ; the existence of *horizontal asymptotes* depends on the behavior of R for large positive and large negative values of x (that is, as $x \rightarrow \pm \infty$). Vertical and horizontal asymptotes will be studied in detail in Chapter 4.

A sketch of the graph of

$$R(x) = \frac{1}{x^2 - 4x + 4} = \frac{1}{(x - 2)^2}; \text{ dom}(R) = \{x : x \neq 2\}$$

is shown in Figure 1.6.5. Note that $x = 2$ is a vertical asymptote and the x -axis is a horizontal asymptote.

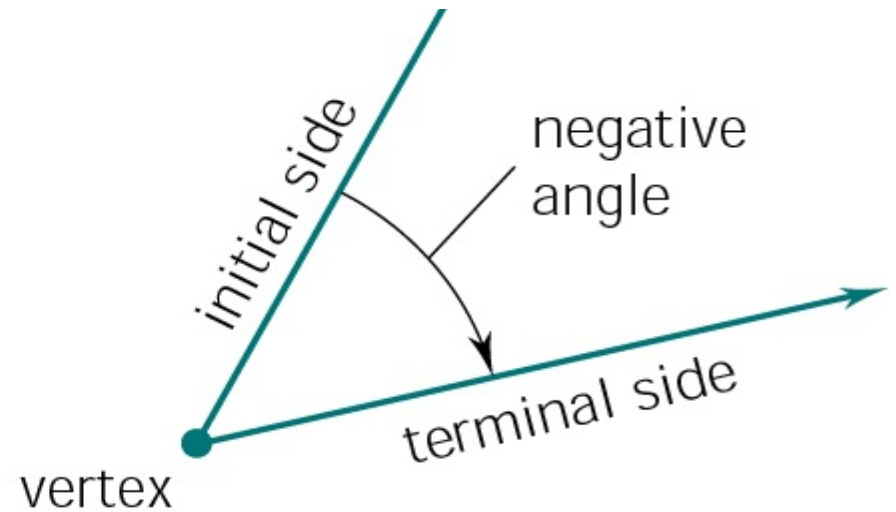
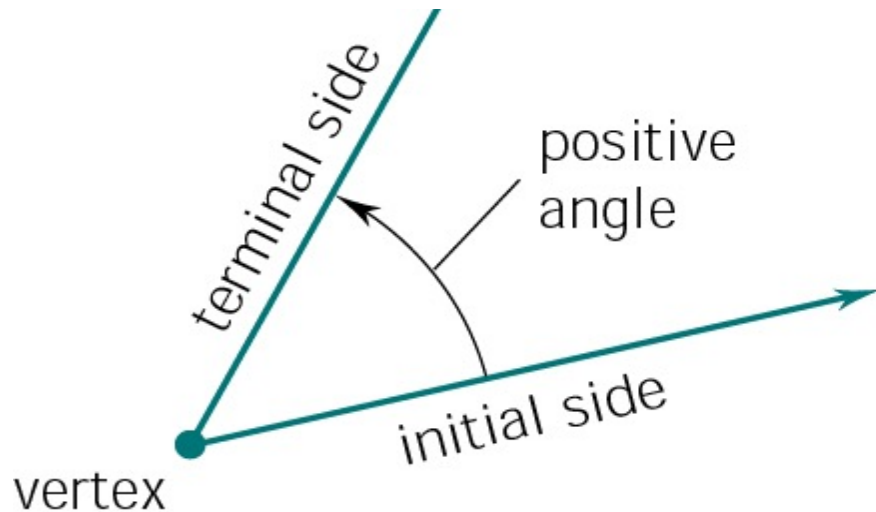


The Elementary Functions

Trigonometric Functions

We assume that the reader is already familiar with trigonometry, and so we present here a brief review of the definitions, basic properties, and graphs of the trigonometric functions.

Angles An angle in the plane is generated by rotating a ray (half-line) about its end-point. The starting position of the ray is called *initial side* of the angle, and the final position of the ray is called the *terminal side*. The point of intersection of the initial and terminal sides is called the *vertex*. An angle is said to be *positive* if it is generated by a counterclockwise rotation and *negative* if the rotation is clockwise (see Figure 1.6.7).

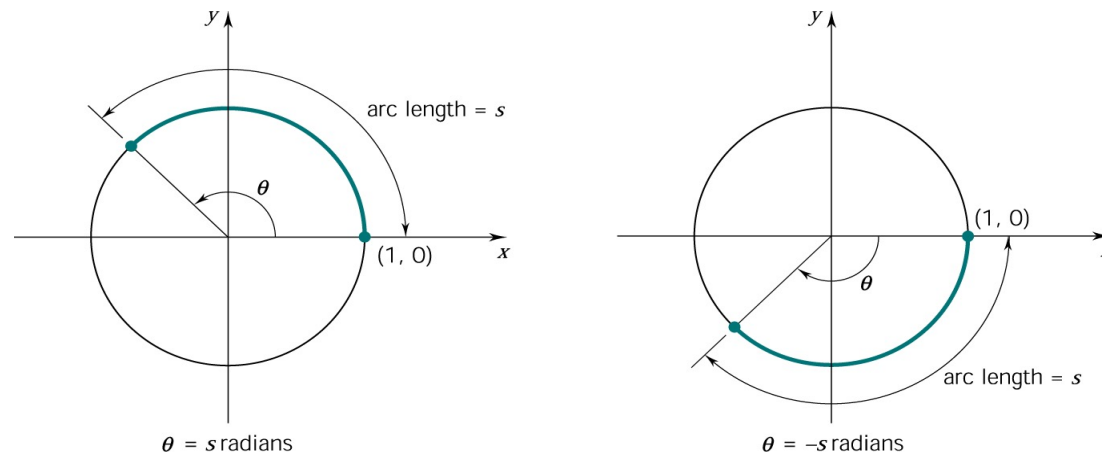


The Elementary Functions

Degree measure, traditionally used to measure angles, has a serious drawback. It is artificial; there is no intrinsic connection between a degree and the geometry of a circle. Why choose 360° for one complete revolution? Why not 100° ? or 400° ?

There is another way to measure angles, called *radian measure*, that associates angles with real numbers, and this is the measure that we will use in applying the methods of calculus to the trigonometric functions. Let θ be an angle in the plane. Introduce a coordinate system such that the vertex of θ is at the origin and the initial side is the positive x -axis. In this context, θ is said to be in *standard position*. Note that the initial side intersects the unit circle $x^2 + y^2 = 1$ at the point $P(1, 0)$. As the initial side is rotated onto the terminal side of θ , the point P traces an arc on the unit circle. Let s be the length of that arc. The *radian measure* of θ is s if θ is a positive angle and it is $-s$ if θ is a negative angle. Thus, the radian measure of θ is a real number; it is the signed length of the arc that θ (in standard position) subtends on the unit circle (see Figure 1.6.8 on page 41). If there is no rotation, then the radian measure of θ is 0.

Since the circumference of the unit circle is 2π , the measure of the angle corresponding to one complete revolution in the counterclockwise direction is 2π radians; half a revolution (a straight angle) comprises π radians; and a quarter revolution (a right angle) comprises $\pi/2$ radians (see Figure 1.6.9 on page 41).



The Elementary Functions

The conversion of degrees to radians and vice versa is made by noting that if A is the measure of an angle in degrees and x is the measure of the same angle in radians, then

$$\frac{A}{360} = \frac{x}{2\pi}.$$

In particular,

$$1 \text{ radian} = \frac{360}{2\pi} \text{degrees} \cong 57.30^\circ$$

and
$$1^\circ = \frac{2\pi}{360} \text{radian} \cong 0.0175 \text{ radian}.$$

The following table gives some common angles measured both in degrees and in radians.

degrees	0°	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
radians	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π	$3\pi/2$	2π

Trigonometric functions – Conversion (deg. To rad.; rad. To deg.), p.41, incl. table

The Elementary Functions

Definition of the Trigonometric Functions We start with the unit circle and use radian measure.

Let θ be an angle in standard position and let $P(x, y)$ be the point of intersection of the terminal side of θ and the unit circle (see Figure 1.6.10).

The six trigonometric functions are defined as follows:

$$\text{sine} \quad \sin \theta = y$$

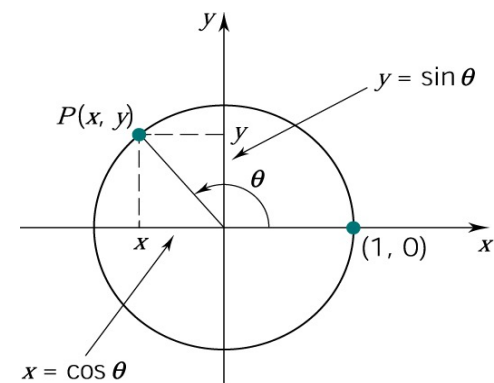
$$\text{cosine} \quad \cos \theta = x$$

$$\text{tangent} \quad \tan \theta = \frac{y}{x} = \frac{\sin \theta}{\cos \theta}, \quad x \neq 0$$

$$\text{cosecant} \quad \csc \theta = \frac{1}{y} = \frac{1}{\sin \theta}, \quad y \neq 0$$

$$\text{secant} \quad \sec \theta = \frac{1}{x} = \frac{1}{\cos \theta}, \quad x \neq 0$$

$$\text{cotangent} \quad \cot \theta = \frac{x}{y} = \frac{\cos \theta}{\sin \theta}, \quad y \neq 0$$



The sine and cosine functions are defined for all real numbers. That is, both sine and cosine have domain $(-\infty, \infty)$. The domains of the tangent and secant functions are all real numbers except for those values of θ for which $x = 0$, namely $\theta = (2n+1)\pi/2$, for any integer n . The domains of the cosecant and cotangent functions are all real numbers except for those values of θ for which $y = 0$, namely $\theta = n\pi$, for any integer n .

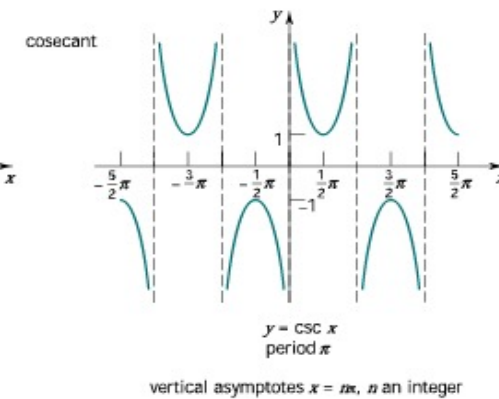
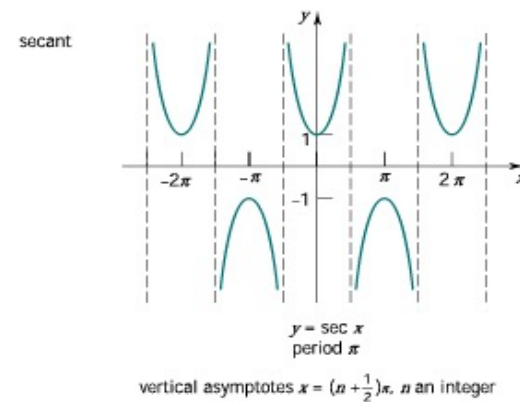
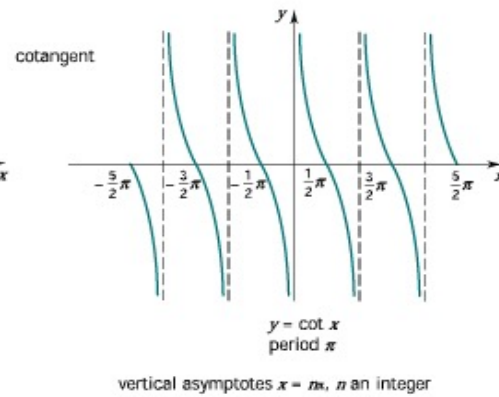
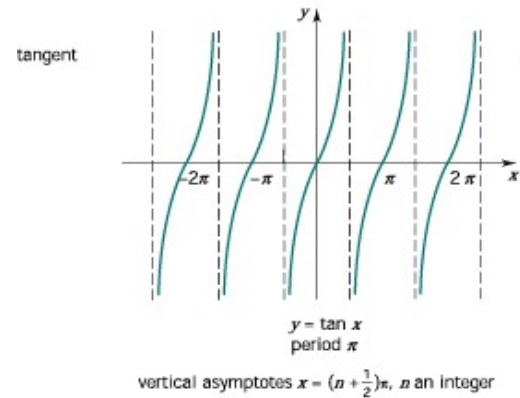
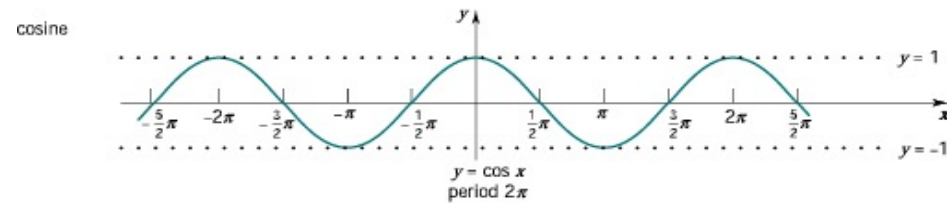
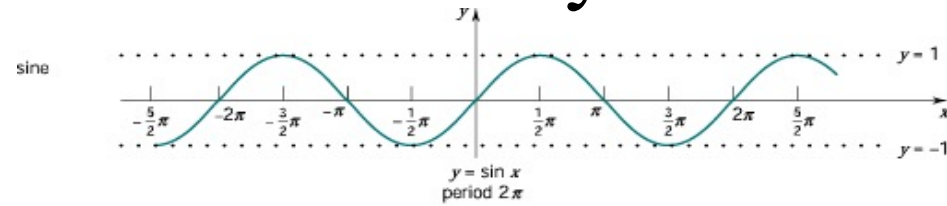
The Elementary Functions

Values We concentrate on the sine, cosine, and tangent functions since cosecant, secant, and cotangent are simply the reciprocals of these functions. The values of the trigonometric functions for the angles $0 (= 0^\circ)$, $\pi/6 (= 30^\circ)$, $\pi/4 (= 45^\circ)$, $\pi/3 (= 60^\circ)$, $\pi/2 (= 90^\circ)$, and selected multiples up to 2π , are given in the following table. You should already have these values memorized.

	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π	$3\pi/2$	2π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1	0	1
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	Undefined	$-\sqrt{3}$	-1	$-\frac{\sqrt{3}}{3}$	0	Undefined	0

In general, the (approximate) values of the trigonometric functions for any angle θ can be obtained with a hand calculator or from a table of values.

The Elementary Functions



Trigonometric functions – Graphs, p.43, 44

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The Elementary Functions

Important Identities

(i) *unit circle*

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \tan^2 \theta + 1 = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

(ii) *addition formulas*

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta,$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta,$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta,$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

(iii) *even/odd functions*

$$\sin(-\theta) = -\sin \theta,$$

$$\cos(-\theta) = \cos \theta.$$

Thus, the sine function is an odd function; its graph is symmetric with respect to the origin. The cosine function is an even function; its graph is symmetric with respect to the y-axis. See Figure 1.6.11.

(iv) *double-angle formulas*

$$\sin 2\theta = 2 \sin \theta \cos \theta,$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

(v) *half-angle formulas*

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta), \quad \cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta).$$

The Elementary Functions

In Terms of a Right Triangle For angles θ between 0 and $\pi/2$, the trigonometric functions can also be defined as ratios of the sides of a right triangle (see Figure 1.6.12).

$$\begin{aligned}\sin \theta &= \frac{\text{opposite side}}{\text{hypotenuse}}, & \csc \theta &= \frac{\text{hypotenuse}}{\text{opposite side}}, \\ \cos \theta &= \frac{\text{adjacent side}}{\text{hypotenuse}}, & \sec \theta &= \frac{\text{hypotenuse}}{\text{adjacent side}}, \\ \tan \theta &= \frac{\text{opposite side}}{\text{adjacent side}}, & \cot \theta &= \frac{\text{adjacent side}}{\text{opposite side}}.\end{aligned}$$

In Exercise 76, you are asked to prove the equivalence of the two definitions.

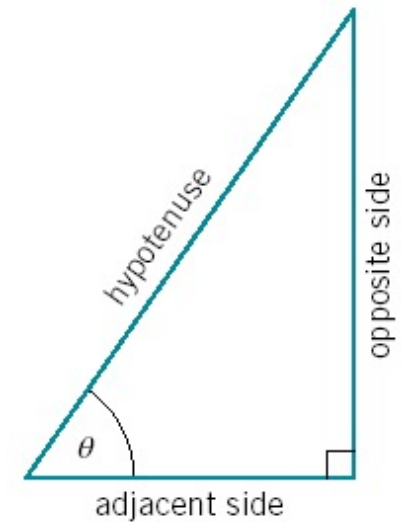


Figure 1.6.12

The Elementary Functions

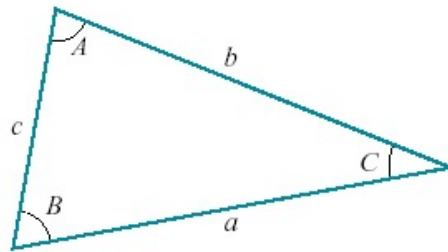


Figure 1.6.13

Arbitrary Triangles Let a , b , and c be the sides of an arbitrary triangle and let A , B , and C be the corresponding angles. (See Figure 1.6.13.)

$$\text{area} \quad \frac{1}{2} ab \sin C = \frac{1}{2} ac \sin B = \frac{1}{2} bc \sin A.$$

$$\text{law of sines} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}.$$

$$\begin{aligned} \text{law of cosines} \quad a^2 &= b^2 + c^2 - 2bc \cos A, \\ b^2 &= a^2 + c^2 - 2ac \cos B, \\ c^2 &= a^2 + b^2 - 2ab \cos C. \end{aligned}$$

Combinations of Functions

Algebraic Combinations of Functions

On the intersection of their domains, functions can be added and subtracted:

$$(f + g)(x) = f(x) + g(x), \quad (f - g)(x) = f(x) - g(x);$$

they can be multiplied:

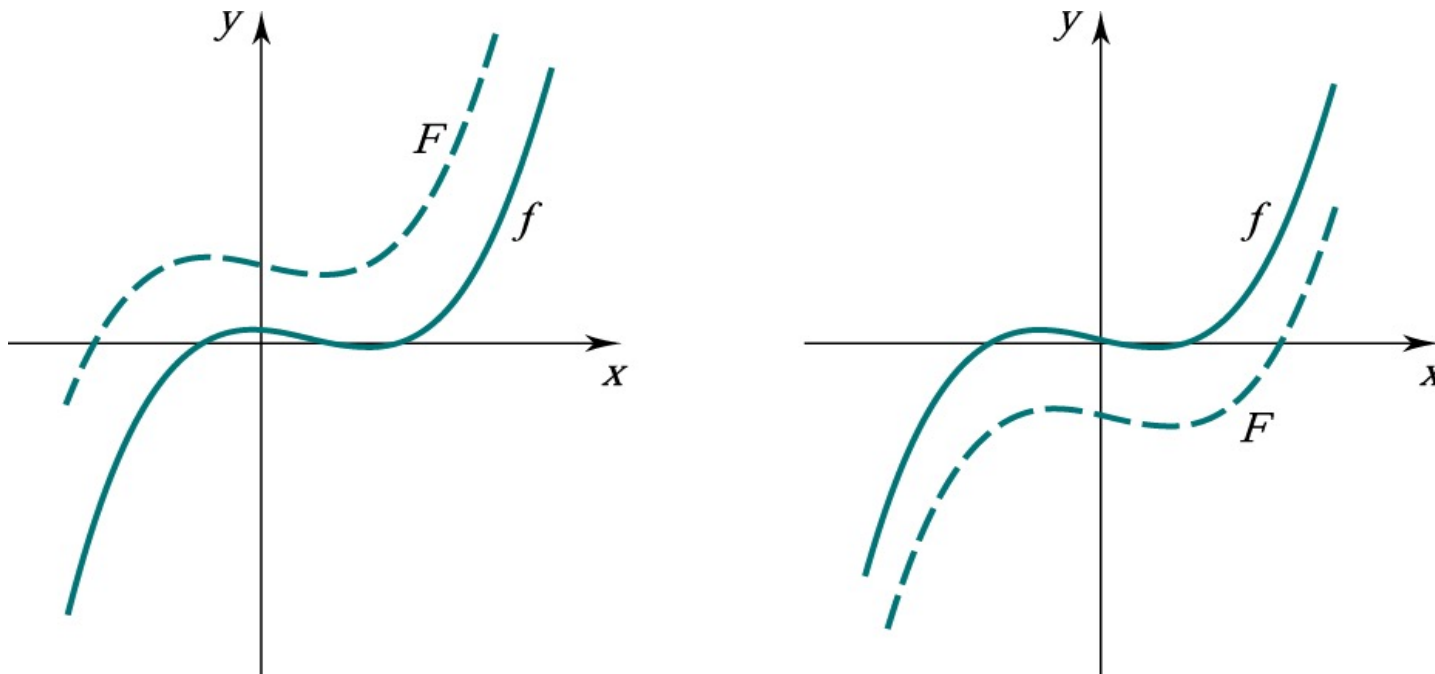
$$(fg)(x) = f(x)g(x);$$

and, at the points x where $g(x) \neq 0$, we can form the quotient:

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Combinations of Functions

There is a special case of addition of functions that is worth noting here. For a function f and a number c , let $F(x) = f(x) + c$ for all $x \in \text{dom}(f)$. The function F is called a *vertical translation* (or *vertical shift*) of f because the graph of F is simply the graph of f shifted up c units if $c > 0$, or shifted down c units if $c < 0$. See Figure 1.7.1.



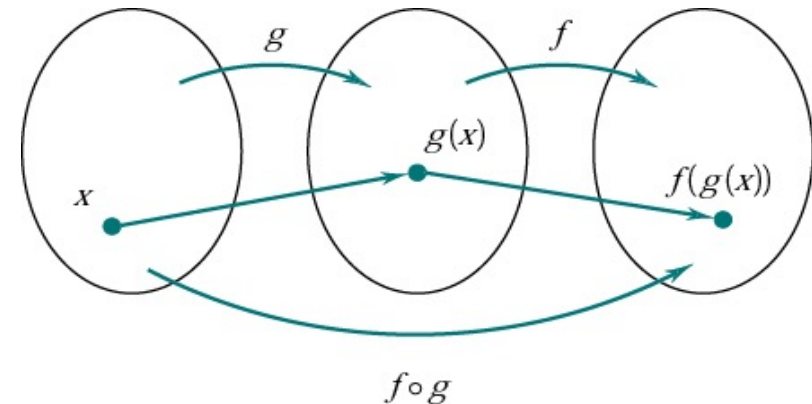
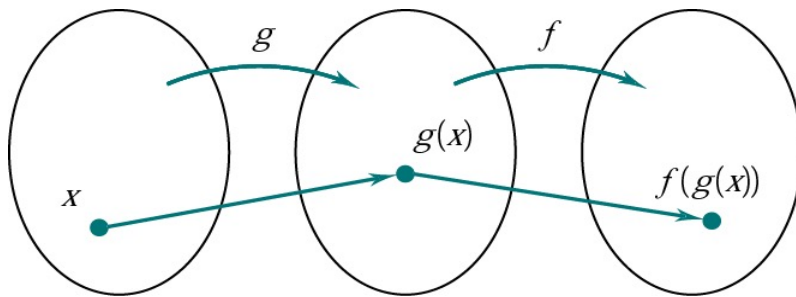
Vertical translation (shift), p.48, figure 1.7.1

Combinations of Functions

Composition of Functions

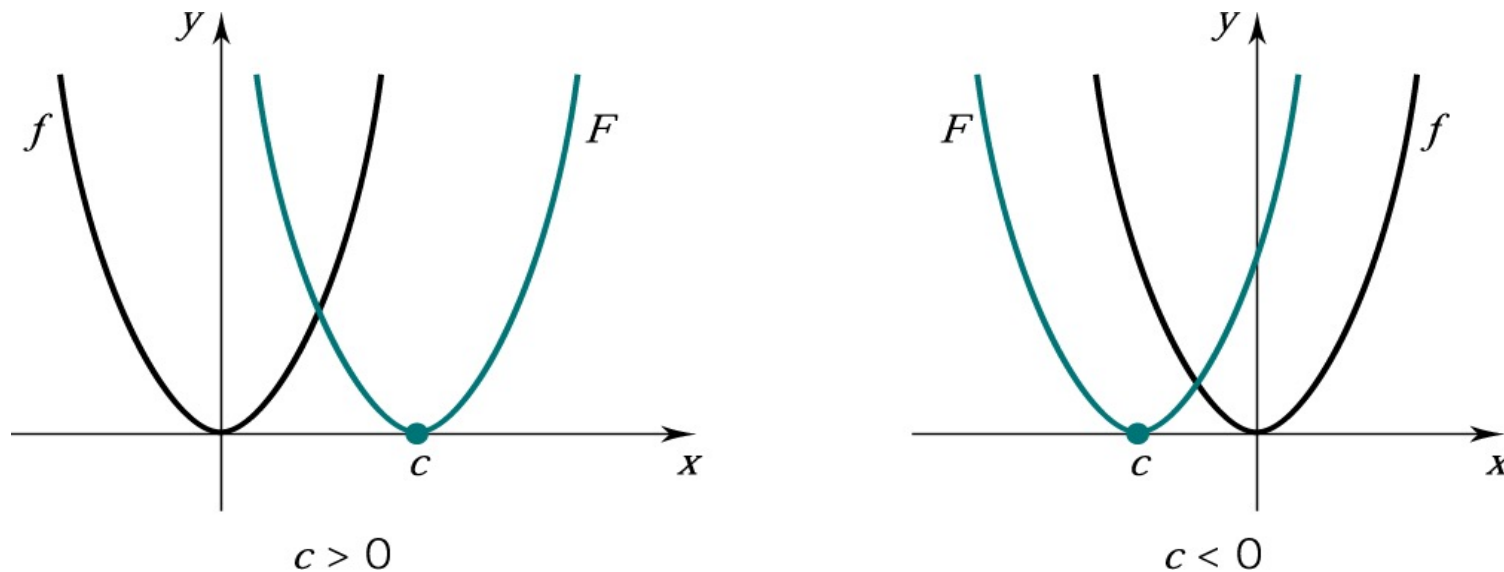
We have seen how to combine functions algebraically. There is another way to combine functions called *composition*. To describe it, we begin with two functions f and g , and a number x in the domain of g . By applying g to x , we get the number $g(x)$. If $g(x)$ is in the domain of f , then we can apply f to $g(x)$ and thereby obtain the number $f(g(x))$.

What is $f(g(x))$? It is the result of first applying g to x and then applying f to $g(x)$. The idea is illustrated in Figure 1.7.3. This new function — it takes x in the domain of g to $g(x)$ in the domain of f , and assigns to it the value $f(g(x))$ — is called the *composition of f with g* and is denoted by the symbol $f \circ g$ (see Figure 1.7.4).



Combinations of Functions

Like addition, there is a special case of composition that is worth noting. For a function f and a number c , let $g(x) = f(x - c)$ for all x in the domain of f such that $x - c$ is also in the domain of f . Then the function g is called a *horizontal translation* (or *horizontal shift*) of f because the graph of g is simply the graph of f shifted c units to the right if $c > 0$, or c units to the left if $c < 0$. See Figure 1.7.5.



Horizontal translation (shift), p.50, figure 1.7.5

Mathematical Induction

Mathematical Induction

Mathematical induction is a method of proof which can be used to show that certain propositions are true for all positive integers n . The method is based on the following axiom:

1.8.1 AXIOM OF INDUCTION

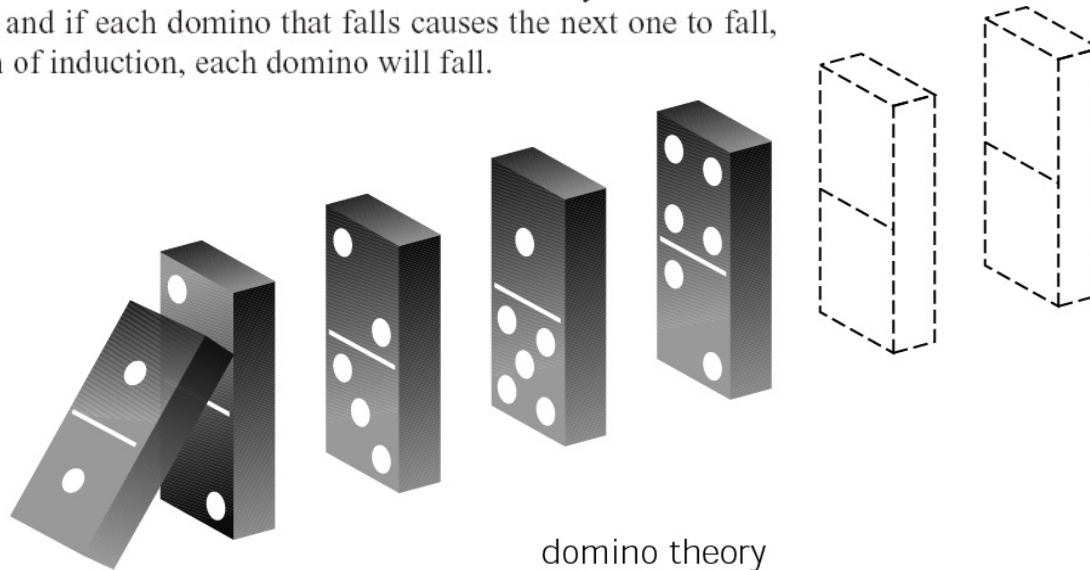
Let S be a set of positive integers. If

(A) $1 \in S$, and

(B) $k \in S$ implies that $k + 1 \in S$,

then all the positive integers are in S .

You can think of the axiom of induction as a kind of “domino theory.” If the first domino falls (Figure 1.8.1), and if each domino that falls causes the next one to fall, then, according to the axiom of induction, each domino will fall.



Axiom of induction, p.55, figure 1.8.1

Exercices

1. Solve the inequality $|3x - 4| < 2$.
2. Proof that $||a| - |b|| \leq |a - b|$.
3. Show that the lines $l_1 : 3x - 4y + 8 = 0$ and $l_2 : 12x - 5y - 12 = 0$ intersect and find their point of intersection.
4. A soft-drink manufacturer wants to fabricate cylindrical cans. The can is to have a volume of 33 ml, remember that 1 ml = 1 cc. Express the total surface area S of the can as a function of the radius and give the domain of the function.
5. Proof by induction that $1 + 2 + 3 + \dots + n = n(n + 1)/2$ for all positive integers n .